MILITARY SPECIFICATION

INDICATOR, AIRSPEED, PITOT STATIC, MAXIMUM ALLOWABLE SPEED, 40-400 KNOTS

This specification has been approved by the Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification covers certain design requirements and all performance requirements for one type of pitot static, maximum allowable airspeed indicator, range 40 to 400 knots.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-P-6906	Plates, Information and Identification
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-P-7936	Preparation for Delivery of Naval Aeronautical Equipment, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys

STANDARDS

Federal

FED. STD. No. 1	Standard for Laboratory Atmospheric Conditions for Testing
FED. STD. No. 595	Colors

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MS25333	Indicator, Airspeed, Pitot Static; Maximum Allowable Speed, 40-400 Knots
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of

FSC 6610

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MS33585	Pointers, Dial, Standard Design of Aircraft
MS33586	Metals, Definition of Dissimilar

Air Force - Navy Aeronautical

AN5727	Glass, Aircraft Instrument Cover

PUBLICATIONS

Air Force - Navy Aeronautical Bulletin

No. 143	Specifications and Standards, Use of
No. 418	Differential Pressures

(When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, 700 Robbins Avenue, Philadelphia 11, Pennsylvania, Attention: Code AD.)

3. REQUIREMENTS

3.1 Qualification. - The indicator furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein.

3.2 Selection of Government documents. - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with ANA Bulletin No. 143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, cotter pins, having suitable properties may be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.2.2 Standard parts. - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials. - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Critical materials. - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Metals. - Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to salt spray or atmospheric conditions to which the indicator may be subjected when in storage or during normal service life.

3.3.2.1 Dissimilar metals. - Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in Standard MS33586 shall not be used in intimate contact with each other.

3.3.3 Nonmagnetic materials. - Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials are essential.

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3.3.4 Fungus-proof materials. - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.5 Corrosive fumes. - The materials used in the indicator shall not liberate deleterious fumes.

3.4 Design and construction. - The indicator shall consist of a pitot static operated indicated airspeed mechanism which will drive a pointer to indicate airspeed on a fixed dial. The indicator shall contain a maximum speed indicating pointer, adjustable on the outside of the case, which may be set to the Mach number setting of the speed at which shock wave phenomena occur in the specific aircraft. The design shall be reasonably simple, to permit service maintenance, repair or overhaul without requiring special tools and fixtures. The indicator shall be constructed to withstand the normal strains, jars, vibrations, and such other conditions as are incident to shipping, storage, installation, and service.

3.4.2 Maintenance. - The design of the indicator shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, and service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.5 Performance. - The indicator shall perform satisfactorily when subjected to the tests specified in Section 4.

3.6 Case. - The outline dimensions of the indicator case shall conform to Standard MS25333. The case shall be pressure tight and shall be so designed that the pressure connection is centered and the static connection is off-center.

3.6.1 Body and mounting flange. - The body and mounting flange of the case shall be made of lightweight nonmagnetic metal, uniform in texture, and shall have a smooth surface. The body shall be finished in a durable black finish and the mounting flange shall be finished with a durable lusterless black finish, Color No. 37038 of FED. STD. No. 595.

3.6.1.1 Insert reinforcement. - The inside of the case around each insert shall be strengthened by a reinforcement to avoid damage to the case when the connection to either insert is tightened during the installation of the indicator.

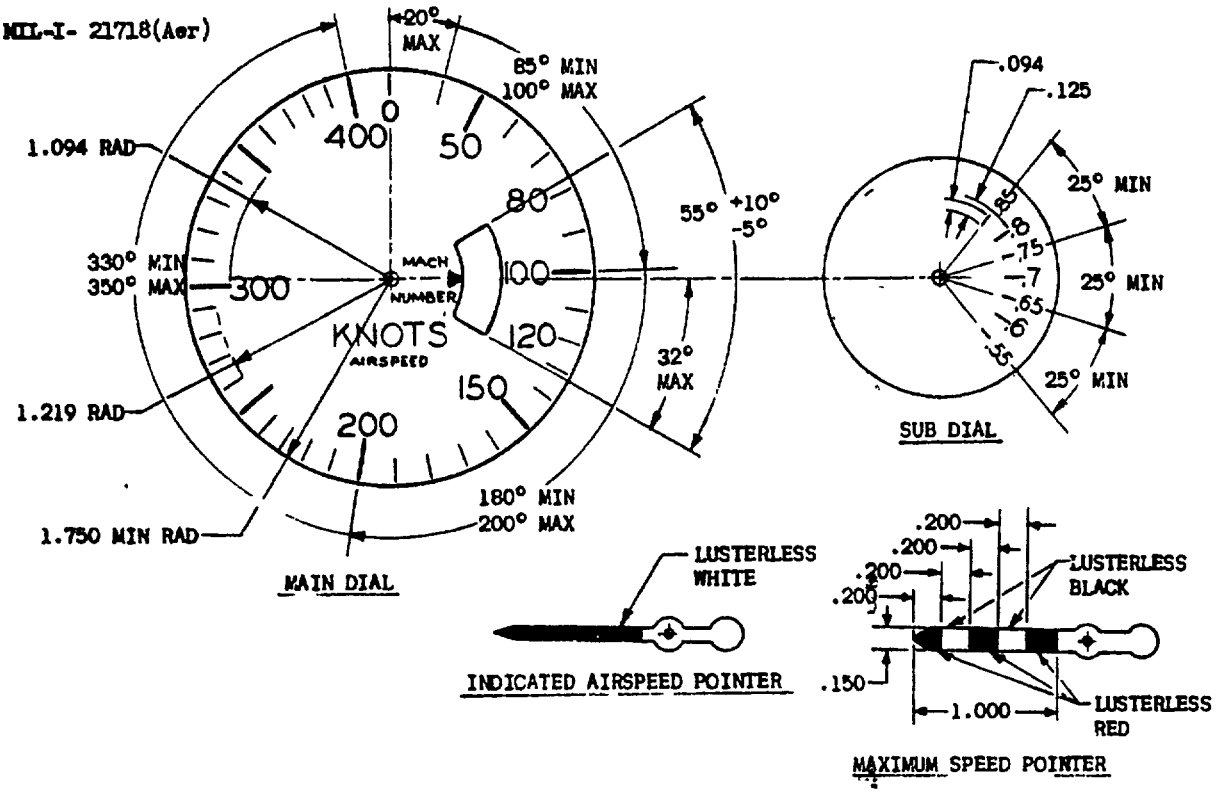
3.6.2 Cover glass. - The cover glass shall be in accordance with Standard AN5727 and shall be properly sealed. It shall be equipped with means to prevent fogging or accumulation of moisture on the surface under extremes of atmospheric conditions.

3.6.3 Dials. - The main dial and the sub-dial shall conform to Figure 1. The dials shall be of aluminum or aluminum alloy and shall be fastened securely into the case or upon the mechanism of the indicator in such a manner that they will not loosen or turn when the indicator is vibrated, except as specified in Section 4. The window in the main dial showing the Mach number setting shall be large enough so that not less than two numerals on the subdial shall be visible at any time.

3.6.4 Dial and cover glass location. - The distance between the front surface of the bezel and the outside surface of the cover glass shall not exceed 0.062 inch. The distance between the inside surface of the cover glass and the main dial shall be the minimum practicable and shall not exceed 0.200 inch.

3.6 5 Pointers. - Two concentrically mounted pointers, conforming to Standard MS33585-3 except as shown in Figure 1 shall be provided. The pointers shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration. They shall be firmly attached to the mechanism and shall be readily adjustable. The maximum speed pointer shall be mounted closest to the dial and the

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ALL GRADUATIONS ARE PARTIAL RADIAL LINES.
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.016

MARKING	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OR GRADUATION ±.005	FINISH
NUMERALS			
0	.156	.016	
50, 80, 100, 120, 150, 200, 300, 400	.188	.031	
GRADUATIONS CORRESPONDING TO			
0	.125	.016	
50, 100, 150, 200, 250, 300, 350, 400	AS SHOWN ON FIGURE	.031	LUSTERLESS WHITE COLOR NO. 37875 OF FED. STD. NO. 595
GRADUATIONS ON MAIN DIAL OTHER THAN THOSE SPECIFIED ABOVE	AS SHOWN ON FIGURE	.016	
LETTERING "KNOTS"	.188	.016	
LETTERING "MACH NUMBER"	.062	.016	
TRIANGULAR MACH NUMBER INDEX	.094	.094	
NUMERALS ON SUB DIAL - .6, .7, .8	.125	.016	
NUMERALS ON SUB DIAL - .55, .65, .75, .85	.094	.016	
GRADUATIONS ON SUB DIAL	AS SHOWN ON FIGURE	.016	
SHADED PORTION OF INDICATED AIRSPEED POINTER	---	---	
LETTERING "AIRSPEED"	.047	.016	LUSTERLESS BLACK COLOR NO. 37038 OF FED. STD. NO. 595
BACKGROUND OF DIALS	---	---	
UNSHADED PORTION OF MAXIMUM SPEED POINTER	---	---	
SHADED PORTION OF MAXIMUM SPEED POINTER	---	---	LUSTERLESS RED COLOR NO. 31136 OF FED. STD. NO. 595

FIGURE 1. DIAL AND POINTER MARKINGS

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indicated airspeed pointer nearest to the cover glass. The nonindicating end of the maximum speed pointer shall not obscure the Mach number scale.

3.6.6 Limitation of pointer movement. -

3.6.6.1 Indicated airspeed pointer. - The indicated airspeed pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 355 degrees of arc, clockwise from the zero graduation, when the indicator is submitted to the Overpressure test. Suitable stops shall also be incorporated in the mechanism to limit the counter-clockwise rotation of the pointer to the zero graduation. The design of the stops shall be subject to the approval of the procuring activity.

3.6.6.2 Maximum speed pointer. - The maximum speed pointer shall be limited by a stop which is adjustable from the front of the indicator by a screwdriver slot in the lower left mounting lug as shown in Standard MS25333. The range of the stop shall be from 150 to 400 knots. The design of this adjustable stop shall be such that it will in no way affect the indication of the pointer when the altitude and Mach number setting are such that the limiting speed will be lower than that set by the adjustable stop.

3.6.7 Mach number adjustment. - The Mach number adjustment shall be made by removing a pipe plug from the rear of the case and adjusting by the use of standard tools. The design of this adjustment shall be subject to the approval of the procuring activity.

3.6.8 Application of pressure. - The pitot pressure shall be applied to the inside of the diaphragm capsule and the static pressure to the interior of the case.

3.6.8.1 Capillary tubing. - A sufficient length of approved capillary tubing having a minimum inside diameter of 0.020 inch, or some other acceptable restriction, shall be incorporated in the indicator case between the pitot connection in the rear of the case and the diaphragm capsule. The effect of this restriction shall be such that the indicator will meet the requirements of the Damping test specified herein.

3.6.8.2 Standard pressure. - The standard pressure used in calibrating the indicator shall be in accordance with the values specified in AMA Bulletin No. 418.

3.6.8.3 Plug. - The static and pitot pressure connections shall each be provided with a vented plug of a material and design acceptable to the procuring activity. The vent hole shall be 0.062 ±0.016 inch in diameter.

3.7 Screw threads. - Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.8 Interchangeability. - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.9 Physical requirements. -

3.9.1 Weight. - The weight of the indicator shall not exceed 1.5 pounds.

3.9.2 Finish. - Protective coatings and finishes which will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

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3.9.2.1 Aluminum alloy parts. - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.9.2.2 Iron and steel parts. - Where practicable, iron and steel parts shall be cadmium, chromium or nickel plated in accordance with Specifications QQ-P-416, QQ-C-320, and QQ-N-290, respectively.

3.10 Marking. - All markings shall be durable to withstand usage encountered in service. The form of the letters and numerals shall conform to Standard MS33558. Numerals shall distinctly indicate the graduation to which each applies.

3.10.1 Dials and pointers. - The dials and pointers shall be marked as shown in Figure 1.

3.10.1.1 Visibility of dial. - The pointers, numerals, at least 1/16 inch of the shortest graduations, and other specified markings on the dial shall be visible from any point within the frustrum of a cone whose side makes an angle of 30 degrees with a perpendicular to the dial and whose small diameter is the aperture of the indicator case.

3.10.2 Case connections. - The back of the case adjacent to the applicable connection shall be marked with the letter "P" for the pressure connection and with the letter "S" for the static connection. The following shall also be marked on the back of the case:

"CAUTION: DO NOT BLOW IN TUBES."

3.10.3 Identification of product. - A nameplate conforming to Specification MIL-P-6906 shall be securely attached to the indicator case, and shall be marked in accordance with Standard MIL-STD-130.

3.11 Installation. -

3.11.1 Installation instructions. - The contractor shall furnish with each indicator one printed copy of instructions with illustrations and diagrams, if necessary, covering the installation of the indicator. Prior to printing, two copies shall be furnished to the Bureau of Aeronautics for approval. The instructions shall be printed on 8-1/2 by 11 or 11 by 17 inch paper.

3.11.2 Mounting screws. - The contractor shall furnish sufficient mounting screws for installing the indicator. The screws shall be No. 6-32 NC-2 fillister head, nonmagnetic, machine screws having a durable and dull black oxidized, or dull black nickel finish. The length shall be sufficient for mounting on a panel up to 3/8 inch thick.

3.11.3 Envelope. - An envelope furnished by the contractor containing installation instructions (and mounting screws if required) shall be packaged with each indicator, and shall be marked with the following information:

"IMPORTANT
THIS ENVELOPE CONTAINS INSTRUCTIONS AND MOUNTING SCREWS"

3.12 Workmanship. - The indicator, including all parts and accessories, shall be constructed and finished to produce an instrument free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of welding and brazing, marking of parts and assemblies, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests. - The inspection and testing of the indicator shall be classified as follows:

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(a) Qualification tests. - Qualification tests are those tests performed on indicators submitted for qualification as a satisfactory product.

(b) Acceptance tests. - Acceptance tests are those tests performed on indicators manufactured and submitted for acceptance under contract.

4.2 Qualification tests. - The qualification tests of the indicator shall consist of all tests of this specification. The tests shall be conducted in the following sequence:

Individual tests
Sampling tests
Altitude
Low temperature exposure
High temperature exposure
Acceleration
Vibration failure
Humidity
Fungus resistance
Salt spray
Mounting lugs
Aging

4.2.1 Qualification test samples. - The qualification test samples shall consist of three indicators of each manufacturer's part number submitted as sample for qualification testing. After obtaining authorization for submittal from the activity responsible for qualification the samples shall be forwarded to the testing laboratory designated in letter of authorization. Each sample shall be plainly identified by a securely attached durable tag marked with the following information: (see 6.3)

Sample for qualification test
Submitted by (name) (date) for qualification tests in accordance with the requirements of Specification MIL-I-21718 under authorization (reference letter authorizing test)
Indicator, Airspeed, Pitot Static, Maximum Allowable Speed, 40-400 Knots
Manufacturer's Part No.
Name of Manufacturer

4.3 Acceptance tests. - The acceptance tests of the indicator shall consist of the individual tests and the sampling tests of this specification.

4.3.1 Individual tests. - The individual tests of the indicator shall consist of the following tests conducted on each indicator:

Inspection
Leakage
Scale errors at room temperature
Hysteresis
After effect
Friction

4.3.2 Sampling tests. - The sampling tests of the indicator shall consist of the following tests conducted on each sampling test sample:

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Magnetic effect
Vibration error
Low temperature operation
High temperature operation
Damping
Position error
Pointer zero position error
Maximum allowable speed pointer stop
Overpressure
Seasoning

4.3.2.1 Sampling test instructions. - Samples shall consist of three indicators selected at random by the inspector from each lot of 100 or less indicators which have passed the Individual tests. A lot shall consist of identical indicators with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. Indicators which have been subjected to the sampling tests shall not be delivered on contract until they have been rebuilt and resubmitted to all Individual tests.

4.4 Rejection and retest. - Indicators which have been rejected as a result of the Individual tests may be reworked to correct the defects, or replaced, and resubmitted for acceptance. Failure of samples representing lots shall be cause for rejection of the lot represented. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units or lots rejected after retest shall not be resubmitted without specific approval of the procuring activity.

4.5 Test conditions. -

4.5.1 Atmospheric conditions. - Unless otherwise specified, all tests required by this specification shall be made in accordance with the requirements of Federal Standard No. 1.

4.5.2 Level position. - Unless otherwise specified, the indicator shall be tested in its level position in which the upper mounting holes are on a horizontal line and the face is vertical.

4.5.3 Maximum speed pointer stop. - The maximum speed pointer stop shall be so adjusted that the maximum speed pointer will not be stopped until it reaches the 400 knot reading.

4.5.4 Tapping. - Unless otherwise specified, the indicator may be gently tapped or vibrated before a test reading is taken.

4.5.5 Vibration stand. - Whenever a vibration stand is specified, it shall be a device which will vibrate at any desired frequency between 5 and 50 cycles per second and shall subject the indicator to such vibration that a point on the vibration stand shall describe, in a plane inclined 45 degrees to the horizontal plane, a circle of the diameter specified herein.

4.5.6 Test points and tolerances. - The indicator test points and tolerances shall be as specified in Tables I, II, and III.

4.6 Test methods. -

4.6.1 Inspection. - Each indicator shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Leakage. -

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TABLE I

TEST VALUES

Tests	Tolerances (Values in knots unless otherwise specified)
Friction:	
Indicated airspeed pointer;	
Between 40 and 200 knots	3.5
Between 201 and 400 knots	5.0
Maximum allowable speed pointer	3.0
Damping:	
Time required from full scale deflection to 50 knots	1.0 $^{+0.5}_{-0.2}$ second
Position Error:	
Indicated airspeed pointer	2.5
Maximum allowable speed pointer	3.0
Pointer Zero Position:	
Indication at 0	1/16 inch
Vibration Error:	
Test altitude - 20,000 feet	
Indicated airspeed - 250 knots	
Maximum vibration - both pointers	2.0
Maximum variation - both pointers	2.0
Overpressure:	2.5
Pitot pressure connection; 140 inches of water	
Static pressure connection; 6.0 inches of water	
(Test altitudes of 0 to 50,000 feet)	
(Test airspeeds of 50 to 400 knots)	
Vibration Failure:	2.0
Test altitude - 20,000 feet	
Indicated airspeed - 250 knots	
Seasoning	2.5
Aging	2.5

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TABLE II

SCALE ERRORS AT ROOM TEMPERATURE (INDICATED AIRSPEED)

Speed Knots	Pressure Inches of Water	Allowable scale errors Knots
40	1.045	5.0
50*	1.634	4.0
60*	2.354	4.0
70	3.207	3.0
80	4.192	2.0
90	5.310	2.0
100*	6.563	2.0
110	7.951	2.0
120	9.475	2.0
130	11.14	2.0
140*	12.94	2.0
150	14.87	2.5
160*	16.95	2.5
170	19.17	2.5
180	21.54	2.5
190	24.05	2.5
200*	26.71	3.0
210	29.51	3.0
220	32.47	3.0
230	35.58	3.0
240	38.84	3.0
250*	42.27	3.0
260	45.85	3.0
270	49.59	3.0
280	53.50	3.0
290	57.58	3.0
300*	61.82	4.0
310	66.24	4.0
320	70.83	4.0
330	75.61	4.0
340	80.56	4.0
350*	85.70	4.0
360	91.04	4.0
370	96.56	4.0
380	102.28	4.0
390	108.20	4.0
400*	114.33	5.0

* Test points.

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TABLE VII

SCALE ERRORS AT ROOM TEMPERATURE
(Maximum Allowable Speed)

Altitude 1,000 Feet	MACH NUMBER .55 Indicated Airspeed Knots	.55 Allowable Scale Errors Knots	.60 Indicated Airspeed Knots	.60 Allowable Scale Errors Knots	.70 Indicated Airspeed Knots	.70 Allowable Scale Errors Knots
0	364*	4	397*	4	--	---
5	334	4	364	4	--	---
10	305*	4	333*	4	390*	4
15	277	4	303	4	356	4
20	251*	4	275*	4	323*	4
25	227	4	248	4	292	4
30	203*	4	223*	4	262*	4
35	181	4	199	4	235	4
40	161*	4	177*	4	209*	4
45	--	--	157*	4	186	4
50	--	--	--	--	165*	4

Altitude 1,000 Feet	.80 Indicated Airspeed Knots	.80 Allowable Scale Errors Knots	.85 Indicated Airspeed Knots	.85 Allowable Scale Errors Knots
0	--	--	--	--
5	--	--	--	--
10	--	--	--	--
15	--	--	--	--
20	373*	4	398*	4
25	337	4	360	4
30	303*	4	325*	4
35	272	4	291	4
40	242*	4	259*	4
45	216	4	231	4
50	192*	4	206*	4

* Test points

4.6.2.1 Case leak. - The pitot and static connections of the indicator shall be joined with a "Y" connection and shall be connected together to the same mercury manometer and sources of suction and pressure. A suction of 15 inches of mercury shall be applied to the pitot and static pressure connections of the indicator with the source disconnected for a period of 10 seconds, the difference between the mercury levels in the manometer shall not change by more than 0.2 inch.

4.6.2.2 Diaphragm capsule leak. - With the static connection open to the atmosphere, a pressure sufficient to produce approximately full scale deflection of the indicated airspeed pointer, shall be applied to the pitot connection of the indicator at which point the connecting tubing shall be pinched off or otherwise completely sealed. During a period of one minute, the indicated airspeed pointer shall not change its position by more than one knot.

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4.6.3 Scale errors at room temperature. -

4.6.3.1 Indicated airspeed. - The indicated airspeed pointer shall be tested for scale errors at the points of the scale indicated by asterisks in Table II. The test shall be made by subjecting the indicator to the pressures specified to produce these readings, first with pressures increasing, then with pressures decreasing. With pressures increasing, the pressure shall be brought up to, but shall not exceed the pressures specified to give the desired reading, and with pressures decreasing, the pressure shall be brought down to, but shall not fall below the pressures specified to give the desired reading. The errors at the test points shall not exceed those specified in Table II. When desired by the Inspector, the indicator shall be tested for scale error at any additional point or points on the scale. The errors at these additional test points shall not exceed those specified in Table II.

4.6.3.2 Maximum allowable speed. - The maximum allowable speed pointer shall be tested for scale errors at the points of the scale indicated by asterisks in Table III. The test shall be accomplished by adjusting Mach number adjustment to each of the settings listed at the top of Table III. During this test the pitot and static connection shall be connected together to a standard mercury barometer and to sources of vacuum and pressure. The reduction in pressure, as indicated by the barometer shall be made at a rate corresponding to an increase in altitude of approximately 3,000 feet per minute. The indicator shall remain at the pressure corresponding to each test point for at least 1/2 minute but not more than 5 minutes before a test reading is taken. The scale errors shall not exceed the allowable scale errors specified in Table III. This test shall be combined at each Mach number setting with the Hysteresis test.

4.6.4 Hysteresis. - Not more than 15 minutes after the indicator has been subjected to the pressure corresponding to the highest altitude specified in Table III, the pressure shall be increased at a rate corresponding to a decrease in altitude of approximately 3,000 feet per minute until the pressure corresponding to 30,000 feet is reached. The indicator shall remain at this pressure for at least 5 minutes, but not more than 15 minutes before the test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until the pressure corresponding to 10,000 feet is reached. The indicator shall remain at this pressure for at least one minute, but not more than 10 minutes, before a test reading is taken. After the reading has been taken, the pressure shall be further increased at the above rate until atmospheric pressure is reached. The reading of the maximum speed pointer at either of the two test points shall not differ from the corresponding reading, with decreasing pressure, by more than 2 knots.

4.6.5 After effect. - Not less than one minute and not more than 5 minutes after the completion of the Scale errors at room temperature (maximum allowable speed) and Hysteresis test, the maximum speed pointer shall have returned to its original reading, corrected for any change in atmospheric pressure, to within 4 knots.

4.6.6 Friction. -

4.6.6.1 Indicated airspeed pointer. - The airspeed scale shall be tested for friction at every other one of the points indicated by an asterisk in Table II, beginning with the first asterisk. The pressure shall be increased to bring the indicated airspeed pointer approximately to the desired reading, and then held constant while two readings are taken, the first before the indicator is tapped and the second after the indicator is tapped. The difference between any two such readings shall not exceed the values specified in Table I. The pointer shall move smoothly while the pressure is varied uniformly without vibration of the indicator. This test may be combined with the test for Scale errors at room temperature (Indicated airspeed).

4.6.6.2 Maximum allowable speed pointer. - The indicator shall be tested for friction at every other one of the points marked with an asterisk in Table III beginning with the first asterisk. The pressure shall be decreased to bring the pointer to the desired reading and held constant while the two readings are taken, the first before the indicator is tapped, and the second, after the indicator

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is tapped. The difference between any two readings shall not exceed the values specified in Table I. The pointer shall move smoothly while the pressure is varied uniformly without vibration of the indicator. This test may be combined with the test for Scale errors at room temperature (Maximum allowable speed).

4.6.7 Magnetic effect. - The indicator shall be revolved about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from the bar magnet, and in a vertical plane which is a perpendicular bisector of the North-South axis of the bar magnet. Starting directly under the compass, the indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions the indicator shall be rotated on its horizontal axis until it is in its normal operating position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed one degree.

4.6.8 Vibration error. - The indicator shall be mounted on a vibration stand with a line joining the centers of the two lower mounting holes in a horizontal plane, with the plane of the mounting lugs of the indicator vertical. This test shall be conducted while the test settings specified in Table I are applied. The indicator shall be vibrated with circular motion in a plane inclined 45 degrees to the horizontal plane with a circle diameter between 0.003 and 0.005 of an inch. The frequency of applied vibration shall be varied slowly from 5 to 50 cycles per second. No condition of applied vibration shall cause vibration errors involving pointer or Mach scale vibration or variation in excess of the tolerance specified in Table I.

4.6.9 Low temperature operation (This test may be combined with the Low temperature exposure test at the discretion of the testing agency). - The indicator shall be properly connected except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of -54°C (-65°F) for a period of 4 hours. During the last 15 minutes of the 4-hour period, power shall be applied to the indicator. At the end of the 4-hour period and with the temperature maintained at -54°C (-65°F) the indicator shall be tested for scale errors, as specified in the Scale error at room temperature test (omitting that part of this test pertaining to Hysteresis) except that the test points shall be every other point in Tables II and III marked with an asterisk, beginning with the first asterisk. Total allowable scale errors shall not exceed the allowable errors specified in Tables II and III by more than 2 knots. No damage to any part of the indicator shall result from this test. During and after this test, the maximum speed pointer stop adjustment shall operate properly.

4.6.10 High temperature operation (This test may be combined with the High temperature exposure test at the discretion of the testing agency). - The indicator shall be properly connected except that no power shall be applied. The indicator shall then be subjected to an ambient temperature of 71°C (160°F) for a period of 4 hours. During the last 15 minutes of the 4-hour period power shall be applied to the indicator. At the end of the 4-hour period and with the temperature maintained at 71°C (160°F) the indicator shall be tested for scale errors as specified in the Scale errors at room temperature test (omitting that part of this test pertaining to Hysteresis) except that the test points shall be every other point in Tables II and III marked with an asterisk, beginning with the first asterisk. The errors at the test points shall not exceed those specified in Tables II and III by more than 2 knots. No damage to any part of the indicator shall result from this test. During and after this test, the maximum speed pointer adjustment shall operate properly.

4.6.11 Damping. - Sufficient pressure shall be applied to the pitot connection to produce full scale deflection of the indicated airspeed pointer. The pressure shall suddenly be released. The time required for the indicated airspeed pointer to reach the value specified in Table I shall be equal to the time and within the tolerance specified therein.

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4.6.12 Position error. -

4.6.12.1 Indicated airspeed pointer. - The indicated airspeed pointer reading, taken while the indicator is held in any desired position and while it is being tapped, shall not differ from its reading while held in any other position, by more than the amount specified in Table I. This test shall be made at every other one of the points of the scale, indicated by an asterisk in Table II, beginning with the first asterisk.

4.6.12.2 Maximum allowable speed pointer. - The maximum speed pointer reading, taken while the indicator is held in any desired position and while it is being tapped, shall not differ from its reading when held in any other position by more than the amount specified in Table I. This test shall be made at every other one of the points of the scale indicated by an asterisk in Table III, beginning with the first asterisk.

4.6.13 Pointer zero position error. - The zero position of the indicated airspeed pointer, with no differential pressure applied to the indicator, shall not differ from the tolerance specified in Table I.

4.6.14 Maximum allowable speed pointer stop. - The indicator shall be disconnected from external sources of vacuum or pressure. The stop mechanism shall be tested for the range by turning the maximum speed pointer adjustment screw to determine the limits of adjustment. The pointer adjustment shall operate smoothly through the entire range of the airspeed scale. No damage or loosening in the mechanism shall result from this test.

4.6.15 Overpressure. -

4.6.15.1 The pressure specified in Table I shall be applied to the pitot pressure connection of the indicator for a period of 10 minutes.

4.6.15.2 The pressure specified in Table I shall be applied to the static connection of the indicator for a period of 10 minutes.

4.6.15.3 Not less than 5 minutes following the application of Overpressure tests the indicator shall be tested for scale errors as specified in the Scale errors at room temperature test. The errors at the test points shall not exceed those specified in Tables II and III by more than the values specified in Table I.

4.6.16 Seasoning. - The indicator shall be subjected to 1,000 applications of a differential pressure sufficient to produce approximately full scale deflection. These applications shall be made at a rate not greater than 6 per minute. Not less than 48 hours following this test, the indicator shall be tested for scale errors as specified in the Scale errors at room temperature test. The errors at the test points shall not exceed those specified in Table II by more than the tolerance specified in Table I.

4.6.17 Altitude. - The indicator properly connected with no power applied shall be tested in accordance with Procedure II of Specification MIL-E-5272 except that the chamber shall be reduced to 3.42 inches of mercury (50,000 feet) for a period of one hour. During the last 15 minutes of the one hour period power shall be applied to the indicator. At the end of the one hour period and with the temperature and pressure maintained as specified the indicator shall be subjected to and shall meet the requirements specified in the Individual tests.

4.6.18 Low temperature exposure. - The indicator shall be tested in accordance with Procedure II of Specification MIL-E-5272. At the end of the exposure period and while still at the -54°C (-65°F) temperature, the indicator shall be tested for scale errors as specified in the Scale errors at room temperature test (omitting that part of this test pertaining to Hysteresis), except that the test

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points shall be every other point in Tables II and III marked with an asterisk, beginning with the first asterisk. The total allowable scale errors shall not exceed the allowable scale errors specified in Tables II and III by more than 2 knots. During and after this test, the maximum speed pointer stop adjustment shall operate properly.

4.6.19 High temperature exposure. - The indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272. After the indicator has returned to room temperature it shall be tested for scale errors as specified in the Scale error at room temperature test (omitting that part of this test pertaining to Hysteresis), except that the test points shall be every other point in Tables II and III marked with an asterisk, beginning with the first asterisk. The total allowable scale errors shall not exceed the allowable scale errors specified in Tables II and III by more than 2 knots. During and after this test, the maximum speed pointer stop adjustment shall operate properly.

4.6.20 Acceleration. - The indicator shall be tested in accordance with Procedure II of Specification MIL-E-5272. The indicator shall be subjected to an acceleration of 20 g's with the indicator in each of the following positions:

(a) Normal operating position with the rear of the case toward the center of rotation.

(b) Rotated 90 degrees in azimuth with respect to the original position.

(c) Dial horizontal, with the zero graduation toward the center of rotation.

At the end of the acceleration test the indicator shall meet the requirements specified in the Individual tests.

4.6.21 Vibration failure. - The indicator shall be tested in accordance with Procedure V of Specification MIL-E-5272. This test shall be conducted using the test settings specified in Table I. The instrument indication at the test settings specified shall be determined with pressures increasing. At the end of the vibration period, the indicator shall be subjected to the pressure specified to produce the test readings (with pressure increasing). The change in airspeed and/or Mach number indication shall not exceed the tolerance specified in Table I.

4.6.22 Humidity. - The indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272. Precaution shall be taken to prevent dripping on the indicator.

4.6.23 Fungus resistance. - The indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272, except that the test period shall be 14 days. At the option of the qualifying agency, tests of component parts of the item may be accepted in lieu of, or in addition to, tests of the assembled instrument.

4.6.24 Salt spray. - The indicator shall be tested in accordance with Procedure I of Specification MIL-E-5272. External connections shall be made to the indicator to simulate installed conditions (see 6.2.2). The indicator shall be subjected to and shall meet the requirements specified in the Individual tests.

4.6.25 Mounting lugs. - The indicator case shall be mounted face downward on the movable head of a suitable testing machine with the face of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug and attached to a pull strap in the stationary head of the machine. A load of 175 pounds shall be applied for one minute to each lug in a direction toward the front of the case. The lugs shall withstand the applied load without fracture and there shall be no damage to any part of the indicator.

4.6.26 Aging. - The indicator shall be retained for a period of not less than 30 days after having been subjected to the Seasoning test, and then tested for Scale errors at room temperature. The scale errors in this test shall not differ from those obtained 48 hours after the 1,000 amplifications of differential pressure specified in the Seasoning test by more than the value specified in Table I.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 Level A. - The indicator shall be packaged in accordance with Specification MIL-P-7936. The method of preservation shall be in accordance with Specification MIL-P-116, Method IId using metal reusable containers.

5.1.2 Level C. - When this level is required packaging shall be in accordance with standard commercial practice

5.2 Packing. -

5.2.1 Levels A and B. - The indicator shall be packed in accordance with Specification MIL-P-7936. The level or levels required shall be as specified in the invitation for bid or contract.

5.2.2 Level C. - The indicator shall be packed to insure that the shipment arrives in a satisfactory condition at destination. The shipment shall conform to the applicable carriers rules and regulations in effect at the time of shipment.

5.3 Marking. - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

6. NOTES

6.1 Intended use. - The indicator covered by this specification is intended for use on high speed aircraft to give an indication of airspeed and Mach number on a single dial.

6.2 Definitions. -

6.2.1 Scale error. - By the "scale error" is meant the difference between the indication of the instrument and the calibrated airspeed corresponding to the applied differential pressure as specified in AMA Bulletin No. 418.

6.2.2 External connections. - When external connections to simulate installed conditions are called for, they shall consist of 10 feet of 1/4 inch copper or aluminum tubing so coiled that moisture can drain out the open end. Similar coils shall be connected to the pitot and static pressure connections.

6.3 Ordering data. - Requisitions, contracts and orders shall specify the MS part number of the indicator desired, the quantity, and the level of packaging and packing to be furnished.

6.4 Qualification. - With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.4.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Aeronautics, Department of the Navy, Washington 25, D. C.

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